

→ Knapsack Algorithm

① $M=100 \Rightarrow$ max limit/weight:

Item	1	2	3	4	5
Profit	10	40	30	40	50
Weight	20	30	66	40	60
P/w	0.5	0.67	0.46	1	0.83

Arrange P/w in ~~ascending~~ $\downarrow$ order
(descending)

Item	I_4	I_5	I_2	I_1	I_3
Profit	40	50	20	10	30
Weight	40	60	30	20	66
P/w	1	0.83	0.67	0.5	0.46

Initially $SW=0$, $SP=0$ $S=\emptyset$

1st pass,

$$SW = 0 + SW_4 = 0 + 40 = 40$$

$(SW \leq M)$ is $SW \leq 100 \checkmark \therefore SP = 0 + SP_4 = 40$
 $S = \{I_4\}$

2nd pass

$$SW = SW + SW_5 = 40 + 60 = 100$$

$$SW \leq 100 \checkmark \therefore SW = 100$$

$$SP = SP + SP_5 = 40 + 50 = 90$$

$SW \leq M \rightarrow$ Stop.

$\therefore$ Max Optimal Profit = 90.

(Fractional)

Q) For the given data ~~n=3~~ $n=3$ $M=20$
Find Max profit using Knapsack Algorithm

	Item	1	2	3
$v =$	Profit	24	25	15
$w =$	weight	18	15	20
$\rightarrow$	P/w	1.33	1.67	0.75

$\therefore$ Arrange P/w in $\downarrow$ order.

Item	I_2	I_1	I_3
Profit	25	24	15
Weight	15	18	20
P/w	1.67	1.33	0.75

Initially, $sw=0$ $sp=0$ $S=\{2,3\}$

1st pass

$$sw = 0 + sw_2 = 0 + 15 = 15$$

$sw \leq M$ $sw \leq 20$ $\checkmark$ $\therefore sp = 0 + sp_2 = 0 + 25 = 25$
 $S = \{I_2\}$

2nd pass

$$sw = sw + sw_1 = 15 + 18 = 33$$

$sw \leq M$ is $sw \leq 20$ $\times$ $\therefore$ No $\rightarrow$ Not possible

$$\therefore \text{Remaining weight} = M - sw = 20 - 15 = 5$$

net wt = 18 $\therefore \frac{5}{18}$ wt can be taken.

$$\text{Fractional} = \frac{M - sw}{w_i} = \frac{20 - 15}{18} = \frac{5}{18}$$

$$S = \{I_2, \frac{5}{18} I_1\}$$

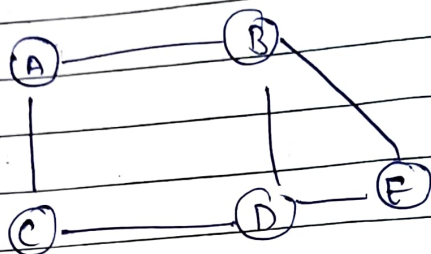
$$sp = sp + sp_1 \times \frac{5}{18} = 25 + \frac{5}{18} \times 24 = 31.67$$

$$sw = sw + sw_1 \times \frac{5}{18} = 20 \therefore \text{optimal Profit} = 31.67$$

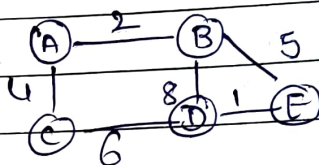
→ MST (Minimum Spanning Tree):

Graph: $G(V, E)$

→ All vertex should be connected to each other.

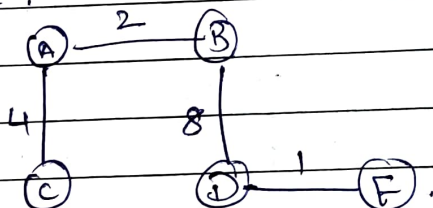


- weighted graph → connected, undirected
 $G(V, E, W)$



- Tree:

→ weighted, connected, no cycles, undirected



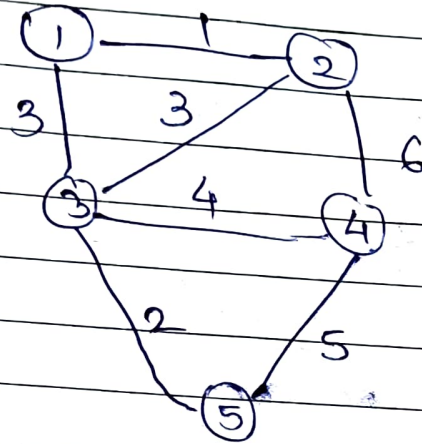
$T(V', E' \subseteq E, W)$

- MST (minimum spanning tree) → connected, weighted, min cost.

MST: A spanning tree with minimum cost, where spanning tree is connected, undirected, (acyclic) and ~~needed~~ weighted that contain all vertices of graph & some or all edges of the graph.

Cond: $V' = V$, $E = V - 1$

Q) For the given ~~one~~ graph find cost of MST using Kruskal's Algorithm.



① Intermediate states not connected

② Check prop:

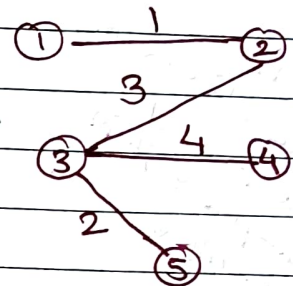
connected, acyclic, weighted. at each stage. show for each step & write reasons also.

→ MST :

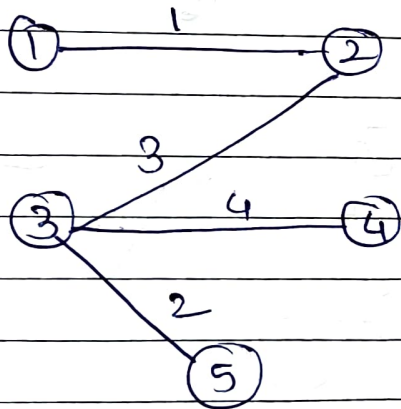
$$V' = V = 5$$

$$E = V - 1 = 4$$

Arrange the cost in $\uparrow$ order.
cost 1, 2, 3, 3, 4, 5, 6.



$$\text{Min. cost} = 1 + 2 + 3 + 4 = 10$$

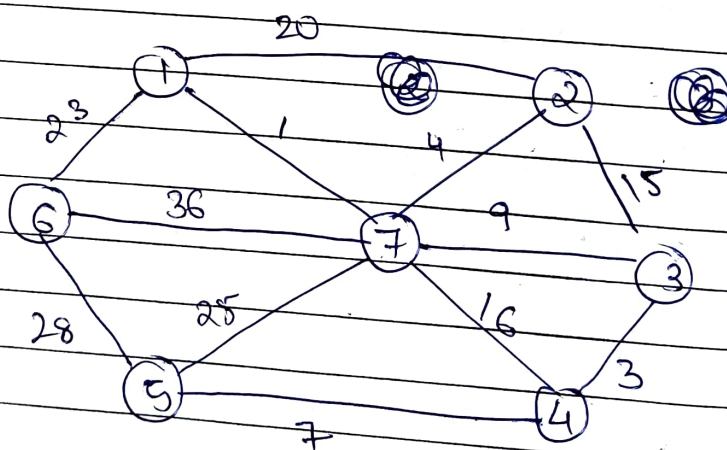
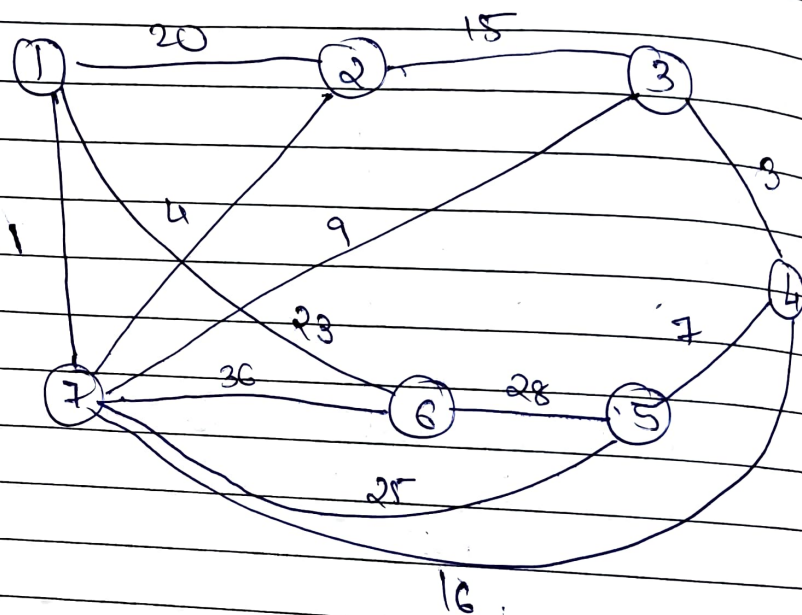


$$\text{Min cost} = 1 + 2 + 3 + 4 = 10$$

Q.)	Edge	Cost	Edge	Cost
	1, 7	1	4, 5	7
	3, 4	3	1, 2	20
	2, 7	4	1, 6	23
	3, 7	9	5, 7	25
	2, 3	15	5, 6	28
	4, 7	16	6, 7	36

Soln.

Step 1:
Draw diagram
the most used
vertex - place it in
the center



cost in $\downarrow$ order.

1, 3, 4, 7, 9, ~~18~~, ~~16~~, ~~20~~, 23, 25, 28, 36

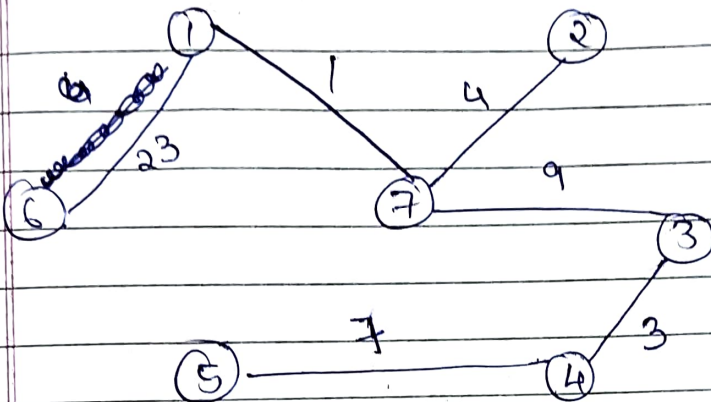
$\therefore$ cycle cycle cycle

$$V' = 7$$

$$E = V - 1 = 6$$

⊗ check prop.

⊗ step by step connect cost in $\downarrow$ order till the no. of edges reqd.



$$\begin{aligned} \text{Min cost} &= 23 + 1 + 4 + 9 + 3 + 7 \\ &= 47 \end{aligned}$$

Step 1: Draw diagram

Step 2: Sort cost in $\downarrow$ order

Step 3: Draw dia with min cost.

$$\begin{aligned} \therefore (1) \text{ Sorting} &= O(n \log n) \\ &= O(e \log e) \end{aligned}$$

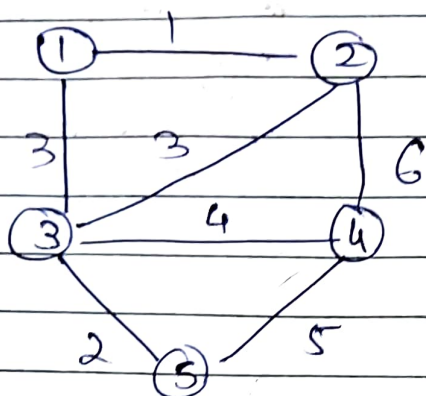
$$\begin{aligned} (2) \text{ selecting} &= O(n) \\ &= O(e) \end{aligned}$$

$$\begin{aligned} \text{Complexity: } &O(e \log e) + O(e) = O(e \log e) \rightarrow \text{worst} \\ \text{best complexity: } &(V-1) \log \frac{V-1}{2} \end{aligned}$$

already
MST

→ Prim's Algorithm

Q) Find cost for given graph using Prim's Algorithm.

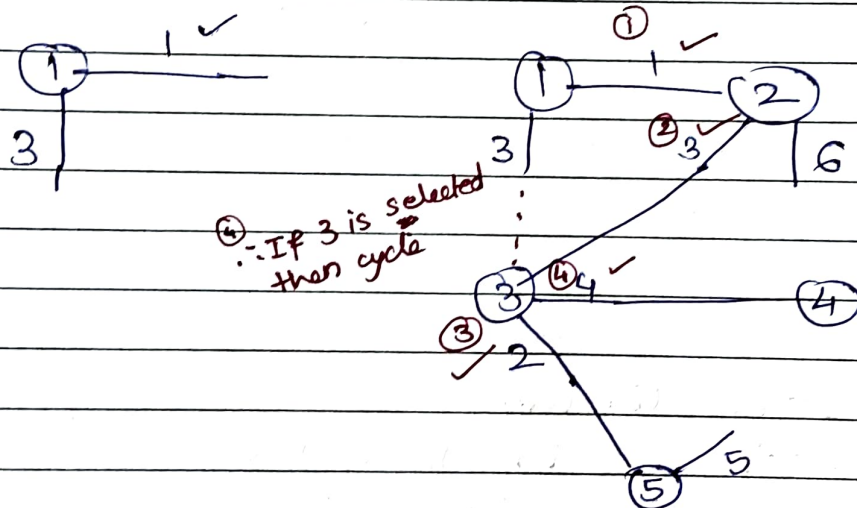


- ① can start from any vertex
- ② check neighbors of that vertex then select 2 fix min path
- ③ Check all the values (edges) left from the next vertex

Soln:

$$V = V' = 5$$

$$E = V - 1 = 4$$



Min cost = 10

Worst case Complexity: $O(n^2)$

Prim's & Kruskal Algo's give same minimum cost.

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